

Cloud Architecture

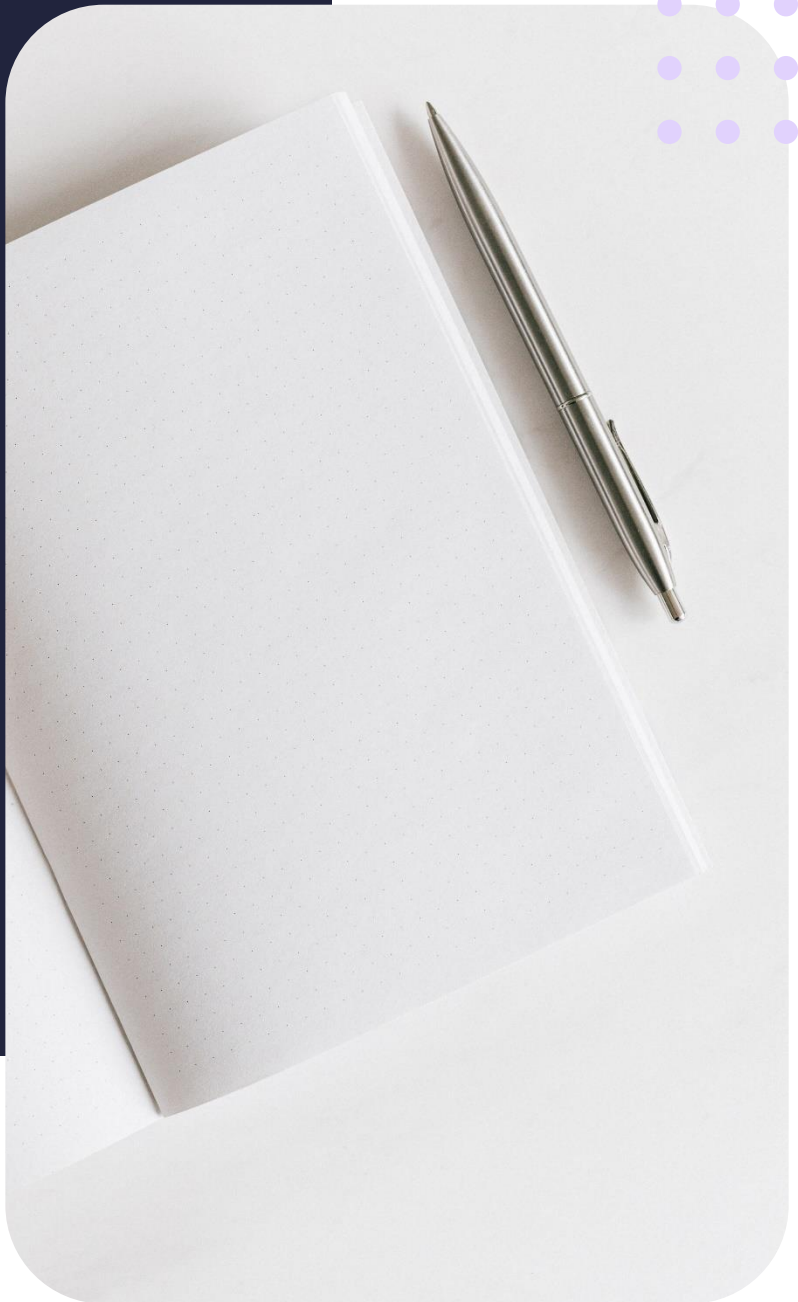
# Oracle APEX: Data analytics

Summary Notes



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## Lesson objectives

By the end of this lesson, you should be able to:

- Discuss the importance of data analytics
- Develop analytical dashboards

## Data analytics

Data analytics is the process of investigating data sets with the use of tools and techniques to identify meaningful trends. It involves using statistical methods, programming, and data visualisation techniques to gain insights and knowledge from structured and unstructured data. The ultimate goal of data analytics is to convert data into actionable insights and inform business strategies.

### Benefits of data analytics

- Provides meaningful insight into data
- Used to optimise business or system efficiency
- Process optimisation may help reduce costs
- Allows businesses to make more informed decisions

### Steps to perform data analytics

1. Identify and validate data requirements
2. Carry out the data collection process
3. Prepare, store and organise data
4. Use tools and techniques to perform analysis

## Types of data analytics

### Descriptive

Descriptive Data Analytics is a type of data analytics that summarises and describes the characteristics of a dataset. It aims to provide a comprehensive overview of the data and highlight key patterns, trends, and relationships. This type of analytics provides a broad picture of the data and helps to identify areas for further investigation.

### Diagnostic

Diagnostic Data Analytics is a type of data analytics that focuses on identifying the root cause of problems or issues through data analysis. It is a process of collecting, analysing, and interpreting data to understand why a particular problem or issue is occurring and to determine the best course of action to resolve it. The main aim of diagnostic data analytics is to determine the underlying causes of problems, rather than simply describing what is happening.

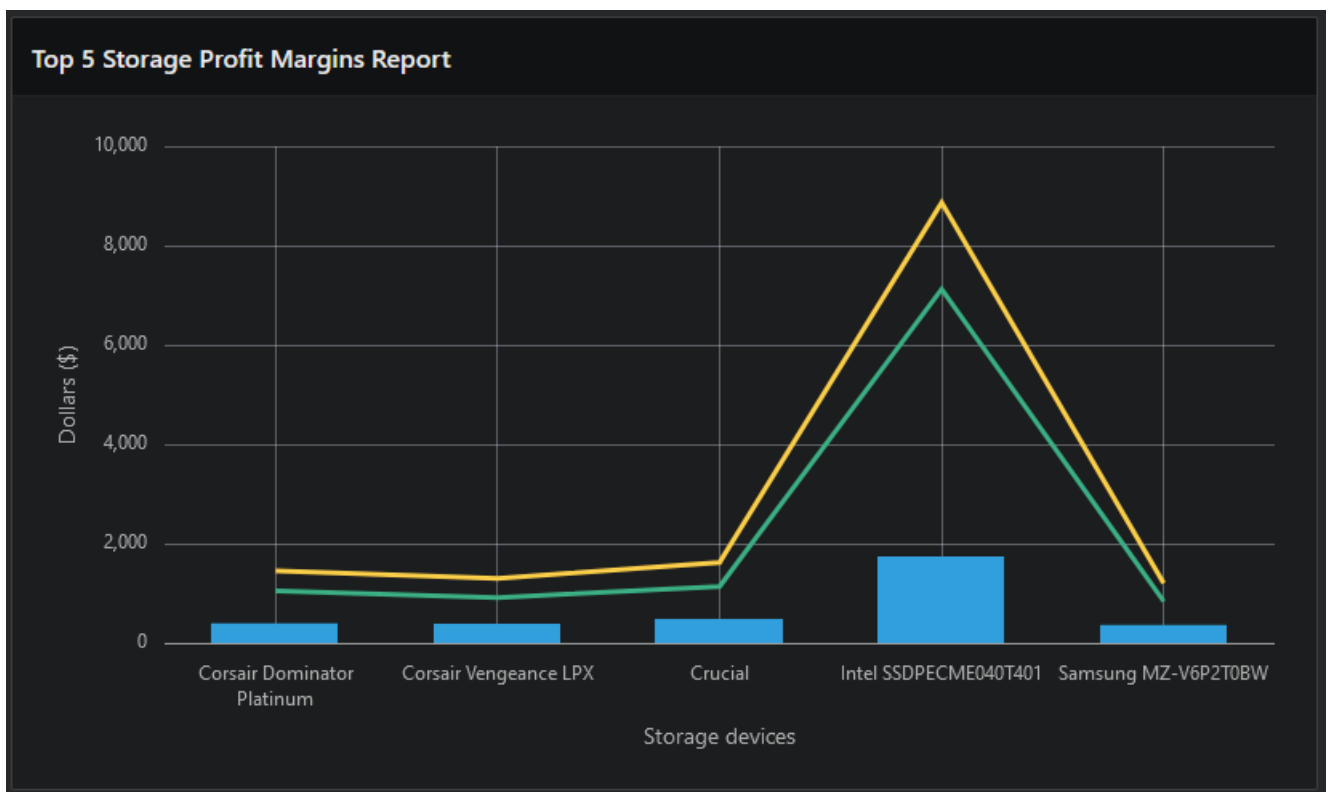
## Predictive

Predictive Data Analytics is a branch of data analytics that focuses on using historical data, statistical algorithms, and machine learning techniques to make predictions about future events. The goal of predictive analytics is to forecast future trends and outcomes based on past data patterns.

## Prescriptive

Prescriptive data analytics is a branch of data analytics that goes beyond descriptive and predictive analytics. It uses optimisation techniques and simulation to determine the best course of action for a given situation. It aims to provide not just what will happen or what has happened, but also what should be done

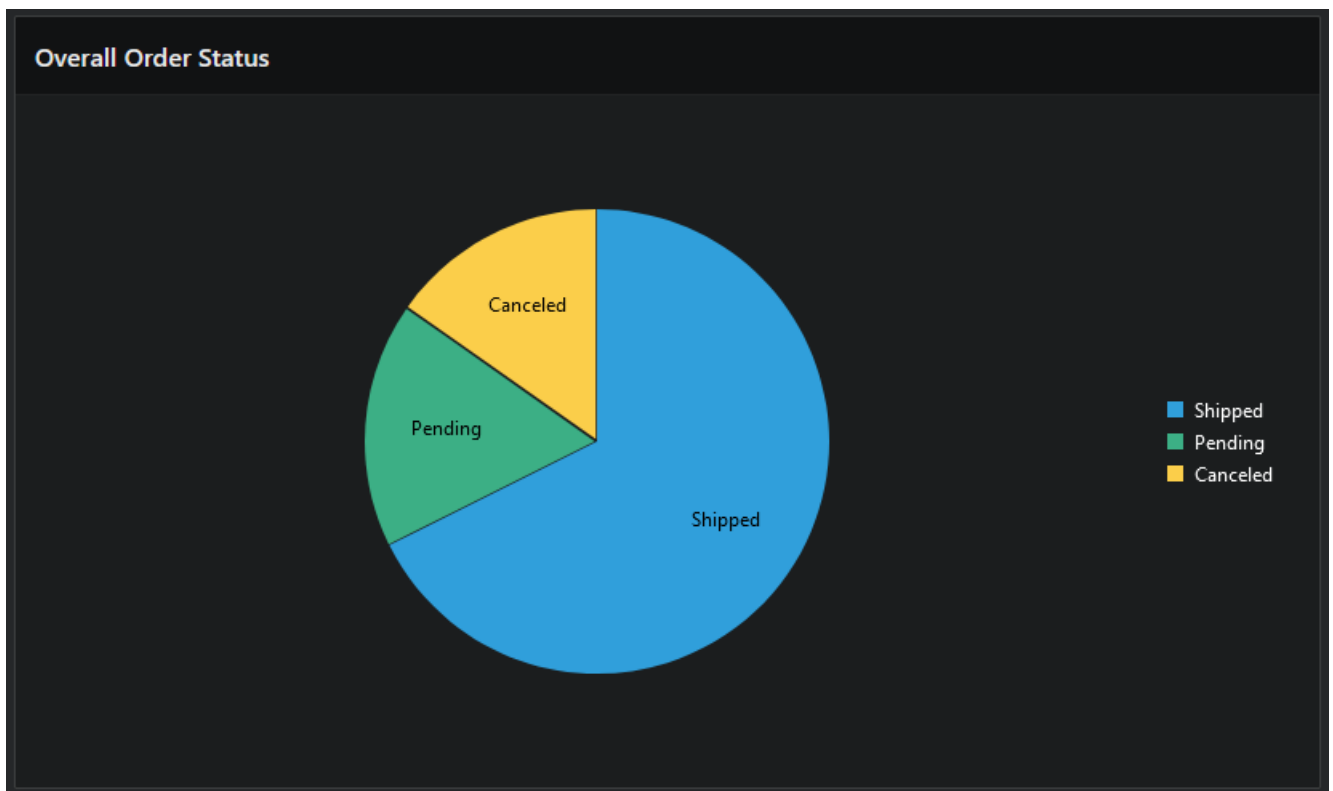
## Top 5 Storage Profit Margins Report



## Source code

```
--SQL query for series data (Profit, Cost and List Price)
select product_name, standard_cost, list_price, category_name, list_price-standard_cost
as profit
from products join product_categories
using (category_id)
where category_id = 5
order by profit desc
fetch first 5 rows only;
```

## Order Status Report



### Source code

```
--SQL query for overall order status pie chart
SELECT COUNT(STATUS) AS "Number", status
FROM ORDERS
group by status
```